

Run report

2026-10-04T035924Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	6	dataset	gaussian
seed	814,786,271	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	1.903×10^{-3}	best_epoch	95
gaussian_params	dim=6, mu_gen=[1, 0, -0.5, 0.5, -1, 0.3], mu_true=[0.8, 0.1, -0.6, 0.7, -0.8, 0.1], cov_gen=[[1, 0.07, 0.22, -0.24, 0, 0], [0.07, 0.49, 0, -0.112, 0.252, 0.098], [0.22, 0, 1.21, 0.088, -0.264, 0.462], [-0.24, -0.112, 0.088, 0.64, 0.096, 0], [0, 0.252, -0.264, 0.096, 1.44, 1.176], [0, 0.098, 0.462, 0, 1.176, 1.96]], cov_true=[[0.64, 0, 0.16, -0.096, 0.08, 0], [0, 0.36, 0, -0.036, 0.12, 0], [0.16, 0, 1, 0, -0.3, 0.44], [-0.096, -0.036, 0, 0.36, 0.12, 0], [0.08, 0.12, -0.3, 0.12, 1, 0.55], [0, 0, 0.44, 0, 0.55, 1.21]], cov_detector=[[0.16, 0, 0, 0, 0, 0], [0, 0.25, 0, 0, 0, 0], [0, 0, 0.36, 0, 0, 0], [0, 0, 0, 0.09, 0, 0], [0, 0, 0, 0, 0.16, 0], [0, 0, 0, 0, 0, 0.16]]		
mmd_sigmas_detector	$2.212 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$1.942 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
parse_config	0.001	0.0	omnifold
data	0.241	0.1	gaussian cache hit – omnifold
omnifold	330.169	98.1	niter=3, epochs=50 – omnifold
init	1.326		DataLoaders and the two MLPs – omnifold
unfold	314.967		MultiFold.Unfold – omnifold
iter1_step1	55.816		detector-level reweighting – omnifold
iter2_step1	68.299		detector-level reweighting – omnifold
iter3_step1	36.918		detector-level reweighting – omnifold
iter1_step2	63.967		particle-level reweighting – omnifold
iter2_step2	55.074		particle-level reweighting – omnifold
iter3_step2	34.856		particle-level reweighting – omnifold
reweight	1.661		evaluating weights on z_target – omnifold
evaluate	6.277	1.9	6 variables – omnifold
total	336.689	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	228.7	9.083	96.0	18.57	91.9	26.59	88.4
dim_1	103.8	25.64	75.3	11.80	88.6	17.30	83.3
dim_2	110.8	16.27	85.3	27.34	75.3	24.60	77.8
dim_3	227.8	7.770	96.6	11.65	94.9	27.44	88.0
dim_4	224.0	6.327	97.2	9.108	95.9	39.13	82.5
dim_5	280.8	7.599	97.3	18.20	93.5	42.85	84.7

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	13.98	0.3119	97.8	0.4107	97.1	0.6830	95.1
dim_1	4.010	0.6668	83.4	0.2743	93.2	0.3153	92.1
dim_2	2.430	0.3828	84.3	0.4144	83.0	0.4164	82.9
dim_3	21.93	0.3663	98.3	0.2972	98.6	0.6797	96.9
dim_4	10.01	0.2561	97.4	0.3877	96.1	0.5148	94.9
dim_5	14.10	0.1335	99.1	0.2251	98.4	0.4766	96.6

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	53.22	1.223	97.7	1.620	97.0	2.673	95.0
dim_1	15.79	2.631	83.3	1.089	93.1	1.251	92.1
dim_2	9.560	1.487	84.4	1.628	83.0	1.630	83.0
dim_3	82.52	1.447	98.2	1.164	98.6	2.657	96.8
dim_4	38.56	1.003	97.4	1.525	96.0	2.029	94.7
dim_5	53.70	0.5136	99.0	0.8823	98.4	1.866	96.5

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	225.4	55.99	75.2	16.12	92.8	30.37	86.5

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	236.0	19.50	91.7	18.50	92.2	25.52	89.2
dim_1	112.7	40.54	64.0	15.17	86.5	18.66	83.4
dim_2	116.6	23.62	79.7	26.14	77.6	23.54	79.8
dim_3	233.6	18.69	92.0	9.852	95.8	28.16	87.9
dim_4	228.2	13.59	94.0	5.738	97.5	35.62	84.4
dim_5	290.1	16.56	94.3	17.38	94.0	43.25	85.1

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	18.34	2.860	84.4	0.3161	98.3	0.7114	96.1
dim_1	8.494	3.475	59.1	0.4917	94.2	0.5808	93.2
dim_2	3.616	1.315	63.6	0.5151	85.8	0.5185	85.7
dim_3	28.54	4.578	84.0	0.2767	99.0	0.9555	96.7
dim_4	12.10	1.453	88.0	0.3064	97.5	0.4613	96.2
dim_5	16.73	1.420	91.5	0.1771	98.9	0.4831	97.1

VLC divergence [$\times 10^3$]

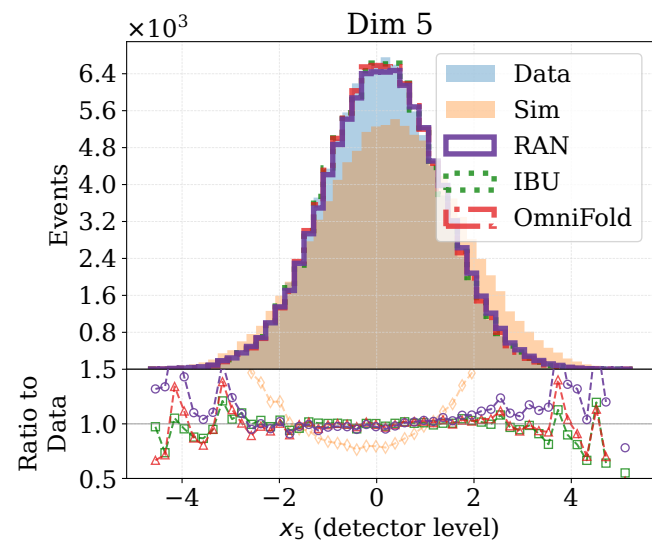
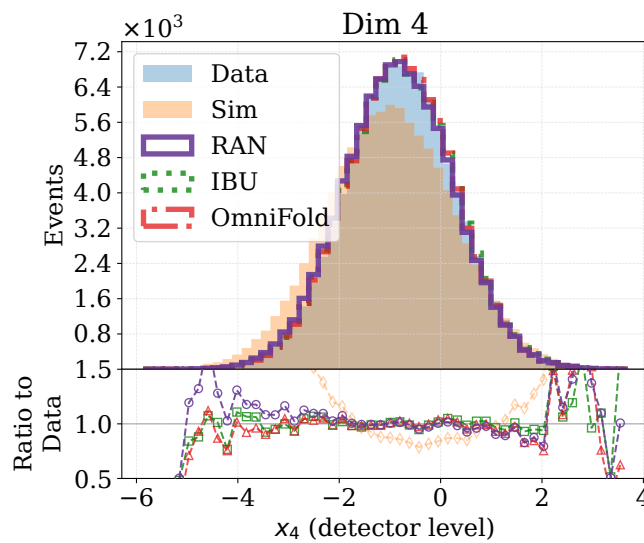
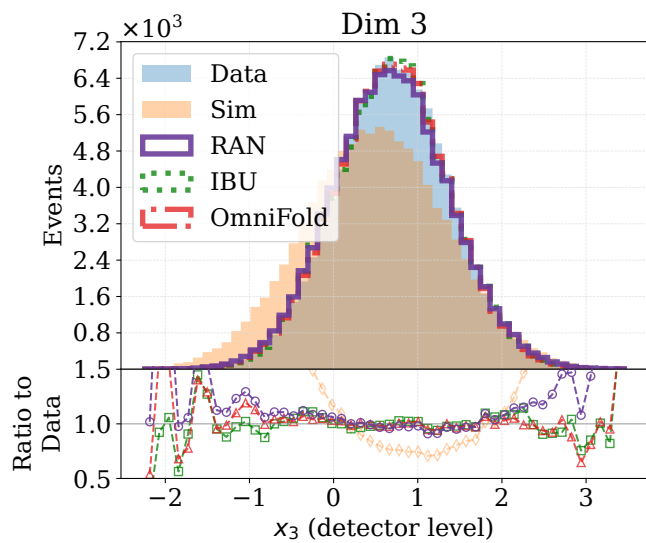
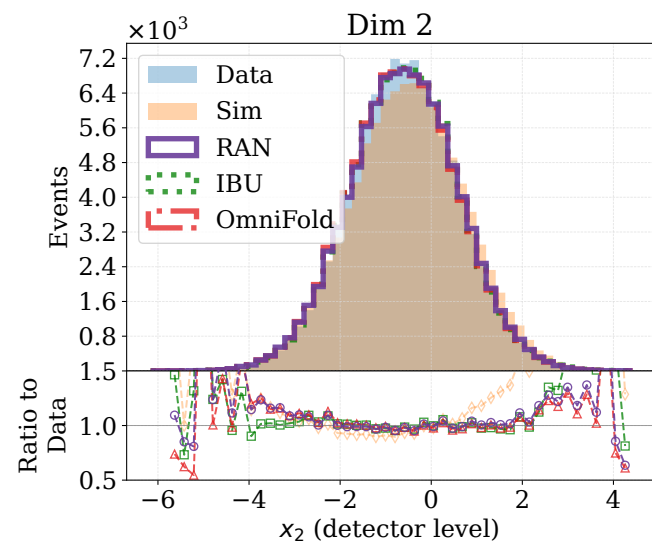
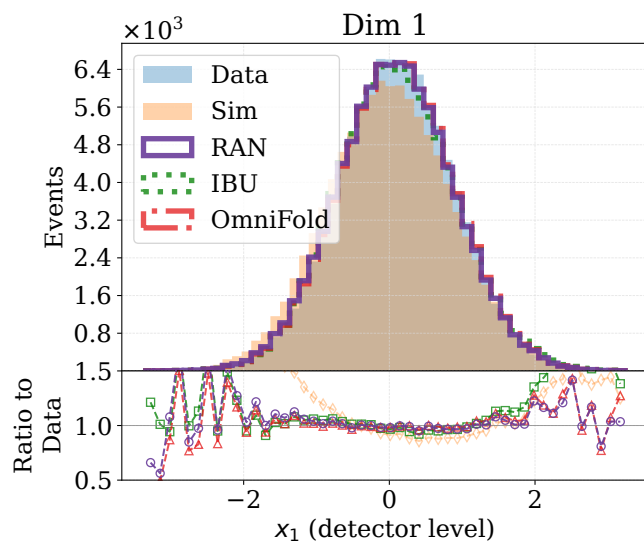
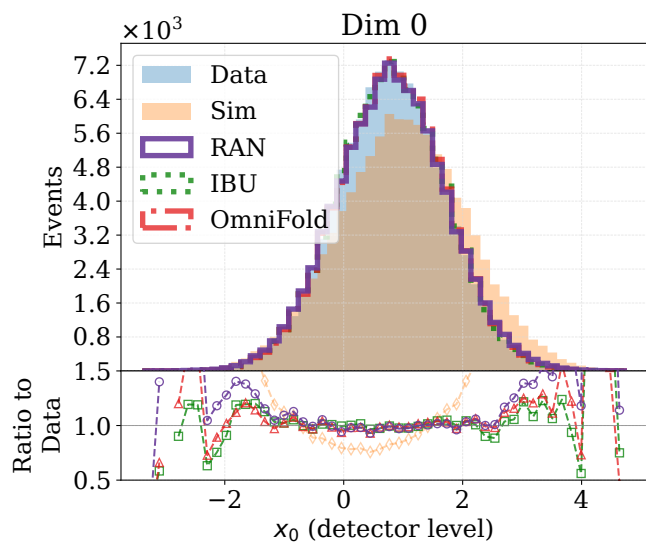
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	68.99	11.33	83.6	1.228	98.2	2.743	96.0
dim_1	32.87	13.70	58.3	1.942	94.1	2.289	93.0
dim_2	14.13	5.163	63.5	2.006	85.8	2.015	85.7
dim_3	105.7	18.11	82.9	1.079	99.0	3.711	96.5
dim_4	46.24	5.759	87.5	1.205	97.4	1.815	96.1
dim_5	63.28	5.640	91.1	0.6903	98.9	1.885	97.0

Sliced Wasserstein, joint [$\times 10^3$]

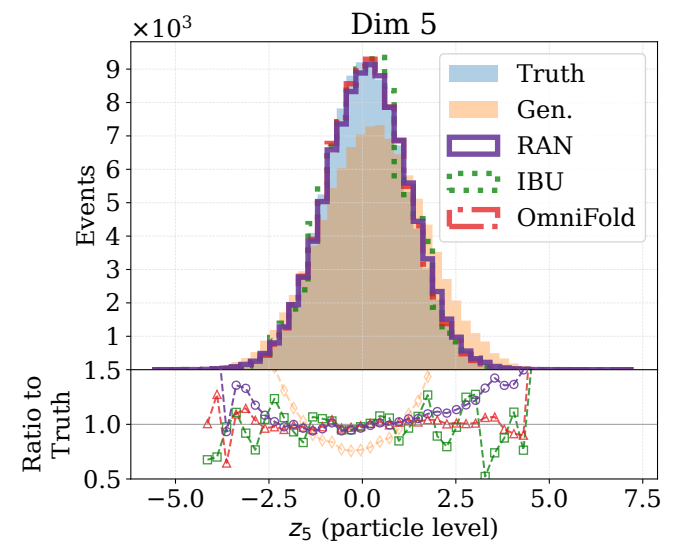
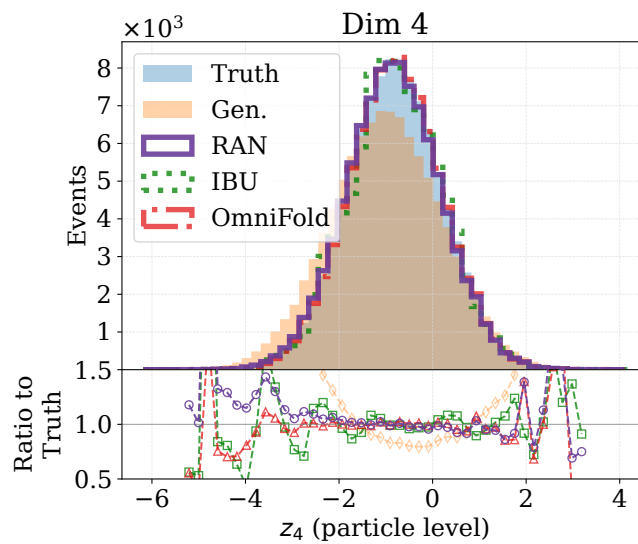
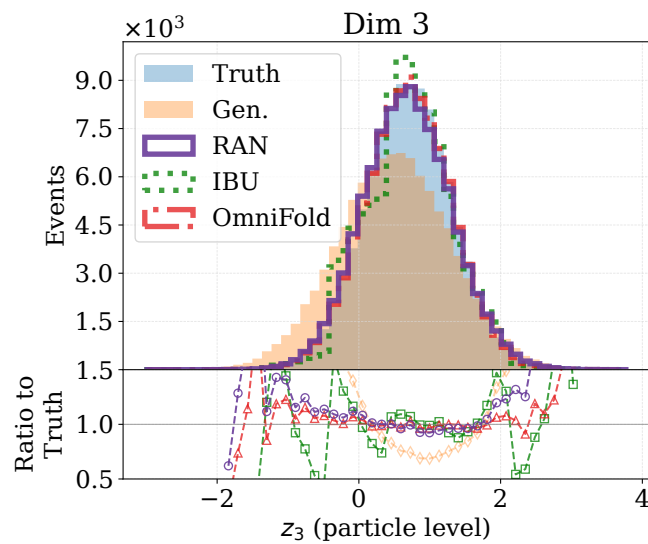
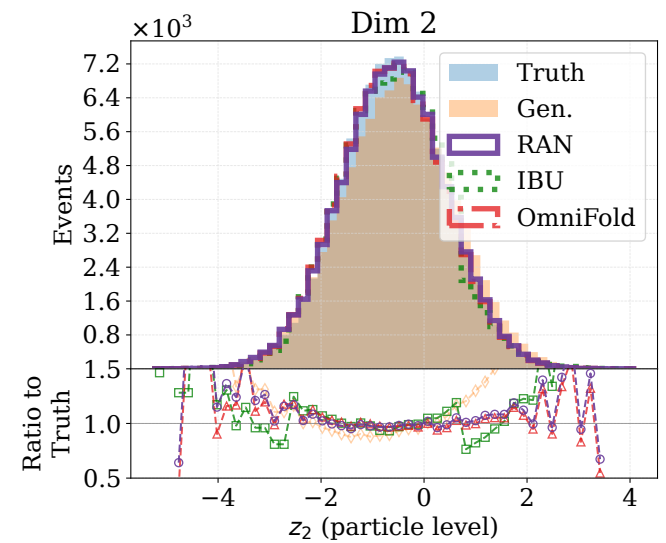
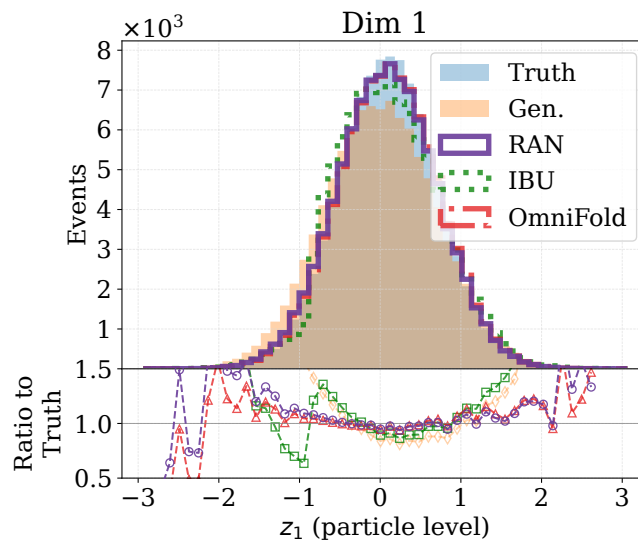
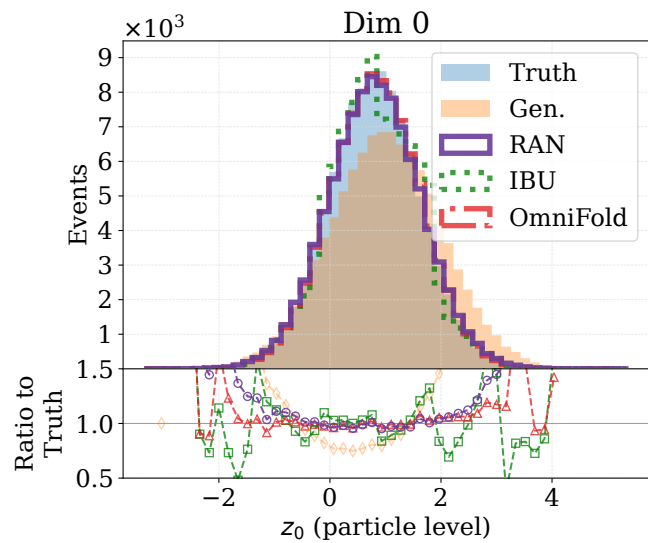
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	261.7	64.71	75.3	17.79	93.2	34.26	86.9

Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

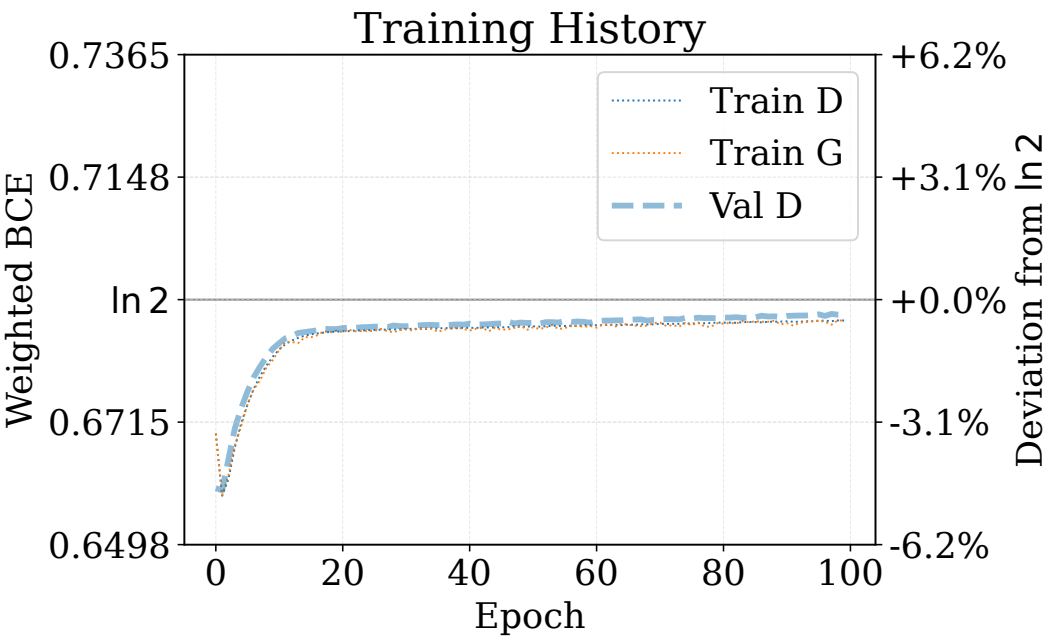
Detector Level



Particle Level



Training diagnostics
Training losses



Selection criteria

